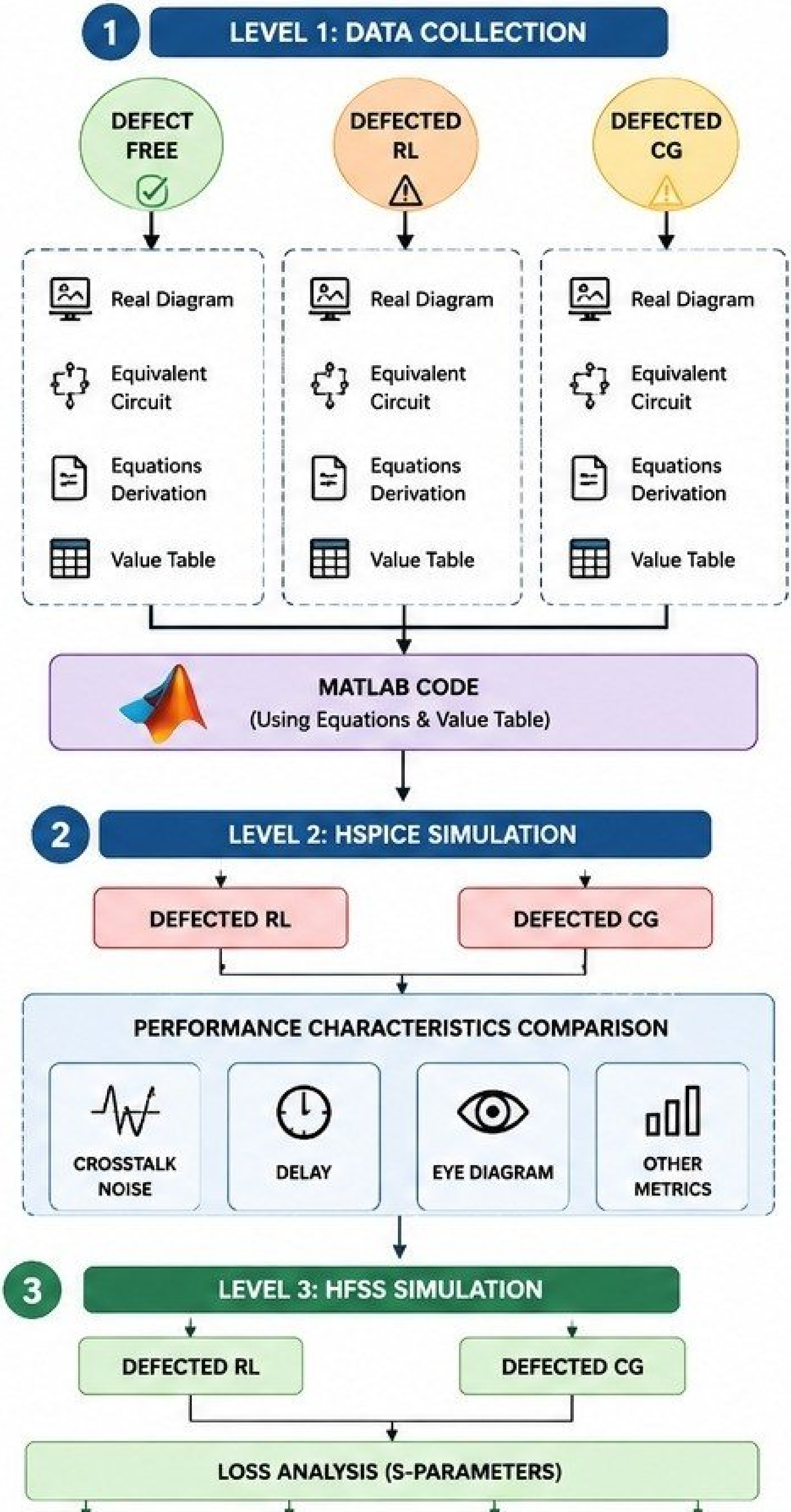
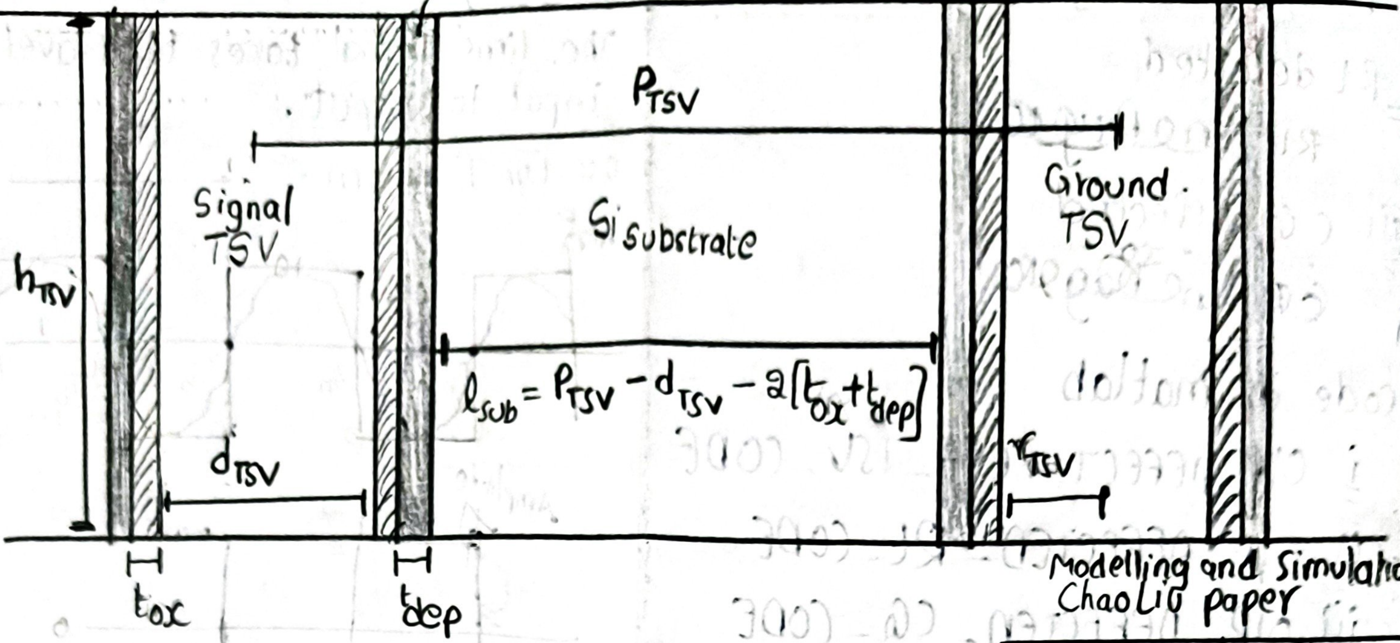
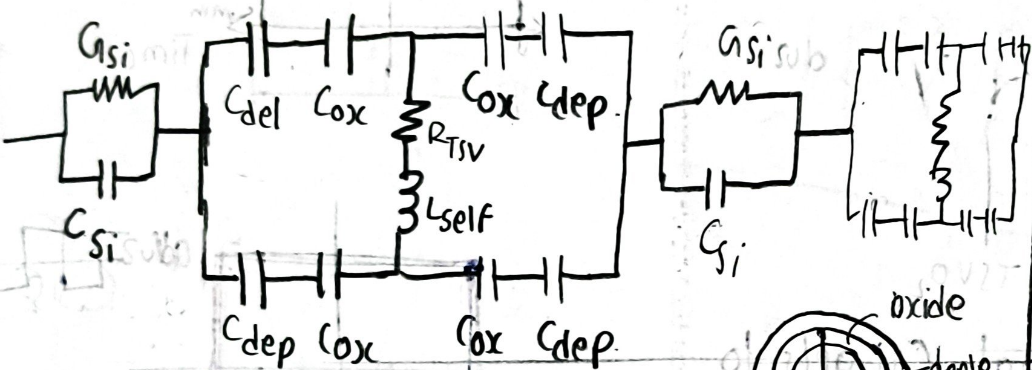

1
LEVEL 1: DATA COLLECTION
DEFECT FREE
DEFECTED RL
DEFECTED CG
Real Diagram
Equivalent Circuit
Equations Derivation
Value Table
Real Diagram
Equivalent Circuit
Equations Derivation
Value Table
Real Diagram
Equivalent Circuit
Equations Derivation
Value Table
MATLAB CODE
(Using Equations & Value Table)
2
LEVEL 2: HSPICE SIMULATION
DEFECTED RL
DEFECTED CG
PERFORMANCE CHARACTERISTICS COMPARISON
CROSSTALK NOISE
DELAY
EYE DIAGRAM
OTHER METRICS
3
LEVEL 3: HFSS SIMULATION
DEFECTED RL
DEFECTED CG
LOSS ANALYSIS (S-PARAMETERS)

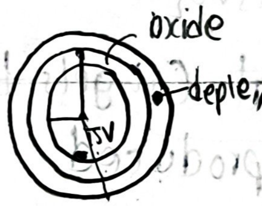
DEFECT FREE (Diagram, Equations, and values).

06/06/26 DAY11 "Kamineni Dhana Sri Keerthi"



Modelling and Simulation Chao Liu paper





Symbol	Value.
$P_{TSV} \cong 4r_{TSV}$	$9.88 \cong 10\,\mu m$
d_{TSV}	$4.94\,\mu m$
r_{TSV}	$2.47\,\mu m$
h_{TSV}	$25\,\mu m$
t_{ox}	$0.13\,\mu m$
t_{dep}	$0.87\,\mu m$
l_{sub}	$3.06\,\mu m$
σ_{si}	$10\ S/m$
σ_{TSV}	$5.9e7\ S/m$
ρ_{TSV}	$1.68\times10^{-8}\,(\Omega m)$
$\varepsilon_{r\,si}$	11.9
$\varepsilon_{r,ox}$	4
$\mu_{r,TSV}$	1
δ_{si}	0
δ_{ox}	0

Equations

1. Resistance.

$$R_{TSV} = \sqrt{(R_{dc\,TSV})^2 + (R_{ac\,TSV})^2} \quad [\Omega]$$

where

$$R_{dc,TSV} = k_p \left[\frac{\rho_{TSV}\, h_{TSV}}{\pi [r_{TSV}]^2} \right] [\Omega]$$

$$R_{ac,TSV} = k_p \left[\frac{\rho_{TSV}\, h_{TSV}}{2\pi r_{TSV}\, \delta_{SD,TSV} - \pi \delta_{SD,TSV}^2} \right] [\Omega]$$

$$\delta_{SD,TSV} = \frac{1}{\sqrt{\pi f\, \mu_{TSV}\, \sigma_{TSV}}} \quad [m]$$

Induc. → 1 self
2 Internal
3 External

2 Inductance.

1 External Inductance of TSV.

$$L_{TSV} = \frac{1}{2}\left[\frac{\mu_0 \mu_{r\,TSV}\, h_{TSV}}{2\pi}\right] \ln\left[\frac{P_{TSV}}{r_{TSV}}\right] \quad [H].$$

∴ Here P_{TSV} (represents distance between current reference to return path.

3 Conductance.

$$G_{Si} = \left[\frac{\pi \sigma_{Si}\,[h_{TSV}]}{\cosh^{-1}\left[\frac{P_{TSV}}{d_{TSV}}\right]}\right] \quad [\text{Siemens}]$$

$$\frac{C_{Si}}{G_{Si}} = \frac{\varepsilon_{Si}}{\sigma_{Si}} \qquad \therefore \varepsilon_{Si} = \varepsilon_0 \varepsilon_{r,Si}$$

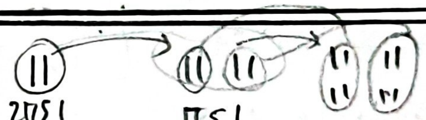
$$\Rightarrow \frac{\left[\frac{\pi \cancel{\varepsilon_0 \varepsilon_{r,Si}}\,[h_{TSV}]}{\cosh^{-1}\left[\frac{P_{TSV}}{d_{TSV}}\right]}\right]}{G_{Si}} = \frac{\cancel{\varepsilon_{Si}}}{\sigma_{Si}}$$

$$V_t = \frac{kT}{q} \simeq 26\,mV$$

$n_i = 1.5 \times 10^{10}$

$N_A = 10$

4 Capacitance.



$$\frac{2\pi\varepsilon l}{\ln\left[\frac{b}{a}\right]} \qquad \frac{\pi\varepsilon l}{\ln\left[\frac{b}{a}\right]} \qquad \frac{1}{2}\frac{\pi\varepsilon l/2}{\ln\left(\frac{b}{a}\right)}$$

(single cylinder TSV)
ln

1 $$C_{ox} = \frac{1}{2}\left[\frac{\pi \varepsilon_0 \varepsilon_{ox}\left[\frac{h_{TSV}}{2}\right]}{\ln\left[\frac{r_{TSV} + t_{ox}}{r_{TSV}}\right]}\right] [F]$$

2 $$C_{dep} = \frac{1}{2}\left[\frac{\pi \varepsilon_0 \varepsilon_{r\,Si}\left[\frac{h_{TSV}}{2}\right]}{\ln\left[\frac{r_{TSV} + t_{ox} + t_{dep}}{r_{TSV} + t_{ox}}\right]}\right] [F].$$

(")
ln.

3 $$C_{Si} = \frac{\pi \varepsilon_0 \varepsilon_{r\,Si}[h_{TSV}]}{\cosh^{-1}\left[\frac{P_{TSV}}{d_{TSV}}\right]} \quad [F]$$

(B/w two cylinders)
$\cosh^{-1}$

$$\Rightarrow t_{dep} = \sqrt{\frac{4 \varepsilon_0 \varepsilon_{r,si} V_{th} \ln\left[\frac{N_A}{n_i}\right]}{q N_A}}$$

where. $\varepsilon_0 = 8.85 \times 10^{-12}$ J/K
$\varepsilon_{rsi} = 11.9$
$V_{th} \cong 25.9mV$
$N_A = 10^{16} cm^{-3}$
$n_i = 1.5 \times 10^{10} cm^{-1}$
$q = 1.602 \times 10^{-19} C$

→ modelling & Simulation by Chao liu' [1]

$$= \sqrt{\frac{4[8.85\times10^{-12}][11.9][0.0259]\left[\ln\left[\frac{10^{15}}{10^{10}}\right]\right]}{(1.602\times10^{-19}][10^{21}]}}$$

$t_{dep} = 0.87\ \mu m.$

$$\Rightarrow l_{sub} = P_{TSV} - d_{TSV} - 2[t_{ox} + t_{dep}]$$

$$= 10 - 4.94 - 2[0.13 + 0.87]$$

$$= 3.06\ \mu m.$$

WORK FLOW [To get Matlab Code values]

i] Defect free	ii] Defected RL	iii] Defected G_{leak} G_{leak}.
✓Real Diagram	Real Diagram	Real Diagram
↓	↓	↓
✓Equ. circuit	Equivalent circuit	Equivalent circuit
↓	↓	↓
✓Equations (RLGC)	Equations (R_{sht}, L_{sht})	Equations (C_{leak} & G_{leak})
↓	↓	↓
✓Value Table.	value table	value table
↓	↓	↓
Equations + values to Matlab code	Equations + values to matlab code	Equations + values to matlab code
↓	↓	↓
Get final values of Equations	Get final values of Equations	Get final values of Eq "
R, L, G, C.	R_{sht} & L_{sht}.	↓
		C_{leak} G_{leak}.